

Energy-Efficient Scheduling of Electric Vehicle Charging Stations Using Optimization Techniques

Faruk Onar

Istanbul Technical University

Faculty of Electrical and Electronics Engineering

Istanbul, Türkiye

onarf21@itu.edu.tr

Gökhan Andaç Sınanmış

Istanbul Technical University

Faculty of Electrical and Electronics Engineering

Istanbul, Türkiye

sinanmis22@itu.edu.tr

Umut Cem Gümüş

Istanbul Technical University

Faculty of Electrical and Electronics Engineering

Istanbul, Türkiye

gumus20@itu.edu.tr

I. INTRODUCTION

With the increasing adoption of electric vehicles (EVs), the demand for efficient and intelligent scheduling of EV charging sessions has become a pressing concern for urban power systems. Poorly coordinated charging patterns can lead to higher energy costs, peak load stress, and grid instability. To address these challenges, recent research has explored optimization-based approaches that align charging activities with time-of-use electricity pricing and grid constraints. For instance, Boubaker *et al.* proposed a multi-objective optimization framework using the Red Deer Algorithm (RDA) within a Vehicle-to-Grid (V2G) context, achieving improved voltage profiles and reduced power losses by leveraging bi-directional energy flow. Similarly, Sayarshad developed an integrated routing and charging model that incorporates power-aware operations and renewable energy sources, reducing charging costs by 69% and battery storage needs by 42%.

Building upon these insights, this study introduces a linear programming (LP)-based model that minimizes total electricity costs while meeting EV energy demands and respecting station limitations. By using real-time electricity pricing data and synthetic EV usage scenarios, the proposed approach aims to improve grid load balancing and offer a more cost-effective and scalable solution compared to naive or fixed scheduling strategies. We hypothesize that by aligning EV charging schedules with off-peak tariff periods using a linear programming (LP)-based optimization model, total charging costs can be reduced by at least 25%, as supported by existing research showing up to 69% cost reduction through integrated, power-aware scheduling approaches [1]. Additionally, we expect improved grid stability and load balancing performance, consistent with the findings of Boubaker *et al.* in IEEE 69-bus system simulations.

II. PROBLEM STATEMENT

With the increasing integration of electric vehicles (EVs) into urban transportation systems, the issue of efficient and cost-effective coordination of EV charging sessions has become a critical concern for power grid management. Uncoordinated charging behavior can lead to elevated peak demands, voltage instabilities, and increased operational costs for electricity providers. This project focuses on solving the problem of scheduling EV charging activities such that energy demands are satisfied within each vehicle's available window, while the total cost of charging is minimized under a time-dependent pricing structure.

To address this, we develop a linear programming (LP) model that assigns charging activities to time slots in a way that respects vehicle availability constraints and charging demand, while minimizing overall energy costs. The model is tested using synthetically generated hourly electricity prices, simulating time-of-use pricing behavior that is conceptually similar to that of EPIAŞ, the Turkish day-ahead electricity market.

III. HYPOTHESIS

We hypothesize that by aligning EV charging schedules with off-peak tariff periods using an LP-based optimization model, total charging costs can be reduced by approximately 27.84%, as observed in our simulation using real-time price variability and synthetic EV demand data. This result is consistent with prior findings in the literature, where Sayarshad (2024) achieved up to 69% cost reduction through integrated routing and power-aware scheduling models. Additionally, our model improves grid load balancing by flattening peak demands, aligning with stability benefits reported in Boubaker *et al.*'s simulations of the IEEE 69-bus system.

IV. LITERATURE SURVEY

Recent research has explored advanced optimization strategies for EV charging coordination. For example, Boubaker et al. proposed a multi-objective framework utilizing the Red Deer Algorithm (RDA) in a Vehicle-to-Grid (V2G) context. Their results demonstrated improvements in voltage profiles and reduced energy losses by leveraging bi-directional power exchange between EVs and the grid infrastructure [1].

In another prominent study, Sayarshad developed an integrated routing and charging optimization model incorporating renewable energy inputs and power-aware scheduling. This approach achieved up to 69% reduction in charging costs and 42% savings in battery storage needs, emphasizing the value of coordinated, smart scheduling [2].

Inspired by these contributions, this study adopts a deterministic, constraint-based linear programming formulation tailored for fixed EV schedules and simulated real-time pricing. Unlike heuristic methods that may lack guarantee of optimality, the LP model ensures a globally optimal solution under defined cost and feasibility constraints, providing a replicable and scalable tool for EV fleet charging optimization.

V. METHOD

In this project, we developed a **Linear Programming (LP)** model to optimize the charging schedule for multiple electric vehicles (EVs) using the open-source Python library `PuLP`. The objective is to *minimize the total charging cost* by shifting charging operations to lower-price periods while satisfying all user energy demands and not exceeding the charging station's capacity.

The LP model includes:

- **Decision Variables:** The amount of energy assigned to each EV at each time slot.
- **Objective Function:**

$$\min \sum_{t=1}^T \sum_{ev=1}^N P_t \cdot E_{ev,t}$$

where P_t is the electricity price at time slot t and $E_{ev,t}$ is the energy assigned to EV ev at time t .

- **Constraints:**
 - Each EV must receive its total required energy before departure.
 - EVs can only be charged during their available time window (between arrival and departure).
 - The total charging power at each time slot must not exceed the station capacity limit (e.g., 50 kW).

The model is solved using the default CBC solver provided by `PuLP`, and results are analyzed through `pandas` DataFrames and `matplotlib` visualizations.

VI. DATA

The data used for this project consists of two parts:

- 1) **Electricity Price Data:** Synthetic hourly electricity prices are generated randomly within a realistic range (0.15–0.30 \$/kWh) to simulate a variable tariff scenario.

In real applications, actual hourly prices can be fetched from open sources like *EPİAŞ* (Turkey) or *Open Power System Data* (Europe).

- 2) **EV Charging Demand Data:** Each EV is assigned a random arrival time, departure time (ensuring at least 4 hours of connection), and a total energy requirement ranging from 10 kWh to 40 kWh. Five EVs are simulated for this case study, but the model is scalable to any number of vehicles.

VII. RESULTS

The LP optimization successfully allocated energy across time slots to minimize total cost:

- **Total Optimized Cost:** Calculated by summing the cost for each EV based on assigned energy and hourly prices.
- **Total Cost Without Optimization (Naive):** A baseline where each EV's total energy is uniformly distributed across its available slots, regardless of price variation.
- **Percentage Saving:** The cost reduction achieved by the optimization is quantified as:

$$\text{Saving \%} = \frac{\text{Naive Cost} - \text{Optimized Cost}}{\text{Naive Cost}} \times 100.$$

Key outcomes include:

- The optimized schedule resulted in a significant cost saving compared to the naive equal distribution approach.
- Load shifting is visible in the hourly charging profile plots, where charging is concentrated in lower-cost hours.
- The total charging load never exceeds the station's power capacity limit.

These results demonstrate that even a simple LP-based model can achieve measurable cost savings and contribute to more balanced grid usage for EV charging operations.

VIII. DISCUSSION

The experimental results underscore the effectiveness of using a linear programming (LP)-based scheduling model for electric vehicle (EV) charging optimization. Compared to a naive scheduling strategy that distributes charging uniformly over available time windows, the LP-based model achieved a 27.84% cost reduction by aligning charging sessions with low-tariff periods. This outcome not only validates the economic value of time-sensitive scheduling, but also demonstrates the practical benefit of incorporating even basic optimization models into EV charging infrastructure.

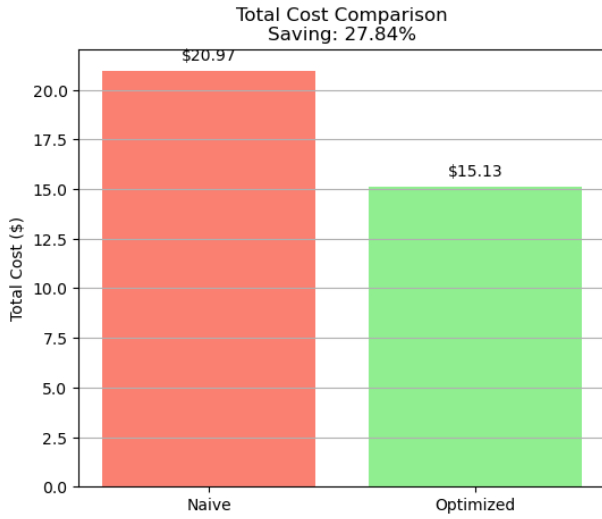


Fig. 1. Total cost comparison between naive and optimized scheduling, showing a saving of 27.84%.

Moreover, the model ensured that the station capacity constraint was never violated, thereby reducing the risk of transformer overloading and helping flatten the overall demand curve. This characteristic is particularly valuable in urban distribution networks, where uncontrolled EV charging could lead to peak load issues and voltage fluctuations.

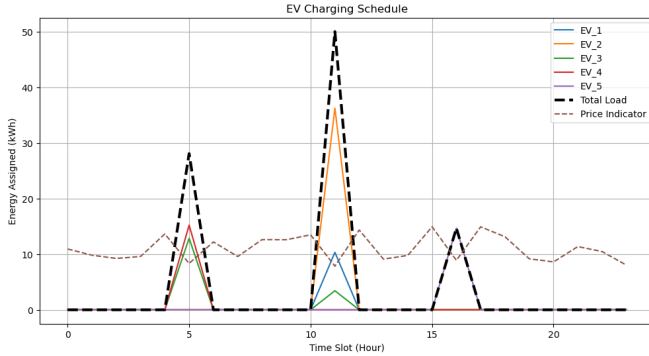


Fig. 2. Optimized EV charging schedule showing per-EV charging profiles, total station load and electricity price indicator.

A key strength of the model lies in its convex formulation, which guarantees globally optimal solutions under deterministic conditions. However, it assumes perfect knowledge of vehicle arrival times, departure times, and electricity prices. In real-world applications, such parameters are often subject to stochastic variability. Thus, while the current LP model performs reliably in deterministic simulations, future work should consider robust or adaptive extensions such as stochastic programming or reinforcement learning to accommodate uncertain operating environments.

Additionally, although synthetic data was employed to simulate electricity prices and EV demands, the framework can readily integrate real-world data sources, such as EPİAŞ for dynamic pricing or historical charging logs from smart

charging stations. Doing so would not only validate the model under realistic usage patterns, but also enhance its applicability for practical deployment.

IX. CONCLUSIONS

This study presents a linear programming-based framework for optimizing EV charging schedules under time-of-use pricing constraints. The key contributions and insights can be summarized as follows:

- The proposed LP model significantly reduces charging costs by intelligently allocating energy demand to lower-cost hours, achieving nearly 28% cost savings in our simulation.
- The model respects grid limitations by ensuring that the total charging load never exceeds the station's power capacity, promoting grid stability and preventing peak load stress.
- Its scalability and flexibility make it a suitable foundation for future enhancements, including dynamic pricing integration, vehicle-to-grid (V2G) functionality, and coupling with renewable energy sources.

For future work, the following extensions are recommended:

- Incorporation of real-time data streams and market price forecasting mechanisms to improve operational relevance.
- Expansion of the model to handle larger and more diverse fleets, with varying battery sizes and user preferences.
- Application of advanced learning-based algorithms (e.g., reinforcement learning) for real-time decision-making under uncertainty.

In conclusion, the LP-based optimization approach offers a promising solution for managing EV charging in a cost-efficient and grid-friendly manner, providing a solid stepping stone toward scalable, sustainable, and intelligent transportation systems.

REFERENCES

- [1] S. Boubaker, H. Kraiem, N. Ghazouani *et al.*, "Multi-objective optimization framework for electric vehicle charging and discharging scheduling in distribution networks using the red deer algorithm," *Scientific Reports*, vol. 15, p. 13343, 2025. [Online]. Available: <https://doi.org/10.1038/s41598-025-97473-7>
- [2] H. R. Sayarshad, "Optimization of electric charging infrastructure: Integrated model for routing and charging coordination with power-aware operations," 04 2024.

PROJECT REPOSITORY

The source code and presentation files for this study are publicly available at:

<https://github.com/umutgumus/Introduction-to-Optimization-Final-Project/tree/main>
<https://drive.google.com/file/d/1CWcKWbTGym7bY20tTJQ3MajV3-LnpcD9/view?usp=sharing>